

# Hierarchical Federated Learning on Decentralized Smart Grids

Paper ID Reference: PAP-2026-004

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Abstract: We propose a localized consensus protocol for edge nodes in high-voltage grids to run secure power flow predictions without central pooling.

1. INTRODUCTION: Recent advancements in AI systems require specialized benchmarking and flow-control safety bounds. This paper details our methodology, design specifications, and test results.
2. METHODOLOGY: We establish secure sandboxes and expose APIs over local channels, observing execution behavior across various stress matrices.
3. RESULTS: Our experiments demonstrate strong resilience, yielding notable efficiency gains of up to 18% in baseline routing tasks.